

Disproof of conjecture 00000007789:
the worst-case log-concave CLT distance d_n is not $O(n^{-1/2})$

Jackmeson1

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1 The conjecture and the clause refuted

Statement (English, verbatim). Definition: The quantitative log-concave CLT: the worst Kolmogorov distance d_n from one-dimensional projections of an isotropic log-concave X to the standard normal, over X and directions. Conjecture: $d_n = O(n^{-1/2})$, the same order as classical independent sums without log factors; and the extremal measure is the generalized simplex (exponential distribution on the simplex), with projecting directions concentrated near the central ray of the simplex, d_n being convex in the deviation from the central direction.

Chinese text. It defines d_n as the worst value (“zui huai zhi”) of the Kolmogorov distance from the law of the projection $\langle X, \theta \rangle$ of an isotropic log-concave X to the standard normal, “ranging over X and θ ” (“bian li X yu θ ”), and conjectures $d_n = O(n^{-1/2})$, followed by the same clauses on the extremal measure.

Reading. Both languages define

$$d_n = \sup \left\{ d_K(\text{Law}\langle X, \theta \rangle, N(0, 1)) : X \text{ isotropic log-concave in } \mathbb{R}^n, \theta \in S^{n-1} \right\},$$

a supremum over X and over directions θ . The conjecture’s later clause, which speaks of the “projecting directions” at which the worst d_n is attained, confirms that the maximum over θ is part of d_n . The conjecture is a conjunction whose first conjunct is

$$(A) \quad d_n = O(n^{-1/2}) \quad (n \rightarrow \infty).$$

We refute (A); this refutes the conjunction. We do not address the later clauses (extremal measure, convexity in the direction), which are not needed.

Reading not addressed. We do not address a “typical direction” quantity (a bound for most θ rather than the supremum over θ), which is a different quantity from the d_n the Definition names. For context, the abstract of B. Klartag, *A Central Limit Theorem for Convex Sets* (arXiv:math/0605014), says: “when the expectation of X is zero, and the covariance of X is the identity matrix, we show that for ‘most’ unit vectors u , the random variable $\langle X, u \rangle$ is distributed approximately according to the gaussian law.”

2 Definitions (as formalized)

All of the following are the definitions in the Lean file `Conjecture7789/Basic.lean`. $\mathbb{R}^n$ is `Fin n → ℝ` with Lebesgue measure (the product of n copies of the 1-D Lebesgue measure).

Definition 1 (log-concave). A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is log-concave (`IsLogConcaveFun`) if it is measurable, $f \geq 0$, and $f(tx + (1-t)y) \geq f(x)^t f(y)^{1-t}$ for all x, y and $0 < t < 1$ (real powers). A measure μ on $\mathbb{R}^n$ is log-concave (`IsLogConcave`) if μ has a log-concave density f with respect to Lebesgue measure: $\mu = \text{volume.withDensity}(\text{ofReal} \circ f)$. (This is the density form of log-concavity. An isotropic measure has identity covariance, so it is not supported on a hyperplane, and the density form is the relevant one.)

Definition 2 (isotropic). μ is isotropic (`IsIsotropic`) if it is a probability measure, every coordinate has a finite second moment (x_i^2 integrable), $\int x_i d\mu = 0$ for all i , and $\int x_i x_j d\mu = \delta_{ij}$ for all i, j (mean 0, covariance I).

Definition 3 (directions, Kolmogorov distance, d_n). A direction is $\theta \in \mathbb{R}^n$ with $\sum_i \theta_i^2 = 1$; the projection is $\langle x, \theta \rangle = \sum_i \theta_i x_i$ (`proj`). For a law ν on $\mathbb{R}$, $d_K(\nu) = \sup_{t \in \mathbb{R}} |F_\nu(t) - \Phi(t)|$ (`kolmogorovToStdNormal`), where F_ν is Mathlib's cdf and Φ is the cdf of `gaussianReal 0 1`. Finally `worstDist n` is the real supremum (`sSup`) of $d_K(\mu \circ \langle \cdot, \theta \rangle^{-1})$ over isotropic log-concave μ on $\mathbb{R}^n$ and directions θ .

Conventions: for $n = 0$ there is no direction, the set is empty and `sSup` gives 0; this does not affect a statement as $n \rightarrow \infty$. The set is bounded above by 1 (each $d_K \leq 1$ because cdfs take values in $[0, 1]$), so the supremum is a genuine least upper bound.

3 Proof

Let $I = [-\sqrt{3}, \sqrt{3}]$, $c = 1/(2\sqrt{3})$, and let $U = c \cdot \text{Leb}|_I$ be the uniform law on I . Let $Q_n = I^n$ be the cube and $\mu_n = U^{\otimes n}$ (`cubeLaw n`).

Lemma 4 (`cubeLaw_logConcave`). μ_n has density $g_n = c^n \mathbf{1}_{Q_n}$, and g_n is log-concave.

Proof. Both μ_n and Leb with density g_n give a box $\prod_i s_i$ the mass $c^n \prod_i \text{Leb}(s_i \cap I)$, so they agree (`cubeLaw_eq_withDensity`, via `Measure.pi_eq`). For $0 < t < 1$: if $x, y \in Q_n$, then $tx + (1-t)y \in Q_n$ by convexity and $g_n(x)^t g_n(y)^{1-t} = c^{n(t+1-t)} = c^n = g_n(tx + (1-t)y)$. If $x \notin Q_n$ then $g_n(x)^t = 0^t = 0$ since $t > 0$; if $y \notin Q_n$ then $g_n(y)^{1-t} = 0$ since $1-t > 0$. In both cases the left side is $0 \leq g_n(tx + (1-t)y)$ (`cubeDensity_logConcave`). $\square$

Lemma 5 (`cubeLaw_isotropic`). μ_n is isotropic.

Proof. $U(\mathbb{R}) = c \cdot 2\sqrt{3} = 1$. $\int y dU = c(\frac{3}{2} - \frac{3}{2}) = 0$ and $\int y^2 dU = c \cdot \frac{(\sqrt{3})^3 - (-\sqrt{3})^3}{3} = c \cdot 2\sqrt{3} = 1$. The coordinate $x \mapsto x_i$ pushes μ_n to U (`measurePreserving_eval`), which gives integrability of x_i^2 , $\int x_i d\mu_n = 0$ and $\int x_i^2 d\mu_n = 1$. For $i \neq j$ the coordinates are independent under the product measure (`iIndepFun_pi`), so $\int x_i x_j d\mu_n = \int x_i d\mu_n \int x_j d\mu_n = 0$. $\square$

Lemma 6 (`cdf_gauss_sqrt3_lt_one`, `cdf_U_sqrt3`). $F_U(\sqrt{3}) = 1$ and $\Phi(\sqrt{3}) < 1$.

Proof. $U((-\infty, \sqrt{3}]) = U(I) = 1$. Lebesgue measure is absolutely continuous with respect to $N(0, 1)$ (`Mathlib gaussianReal_absolutelyContinuous`), and $\text{Leb}((\sqrt{3}, \infty)) = \infty \neq 0$, so $N(0, 1)((\sqrt{3}, \infty)) > 0$, that is, $\Phi(\sqrt{3}) < 1$. $\square$

Theorem 7 (`delta_le_worstDist`). Let $\delta = 1 - \Phi(\sqrt{3}) > 0$ (numerically $\delta \approx 0.0416$). For every $n \geq 1$, $d_n \geq \delta$.

Proof. Take $X \sim \mu_n$ and $\theta = e_1$ (`Pi.single i 1` with $i = 0$), so $\sum_i \theta_i^2 = 1$ and $\langle x, \theta \rangle = x_1$. The law of $\langle X, \theta \rangle$ is U (`map_proj_cubeLaw`). Then $d_K(U) \geq |F_U(\sqrt{3}) - \Phi(\sqrt{3})| = 1 - \Phi(\sqrt{3}) = \delta$, and $d_n \geq d_K(U)$ since (μ_n, e_1) is admissible. $\square$

Theorem 8 (`worstDist_not_isBig0, conjecture_false`). d_n is not $O(n^{-1/2})$ as $n \rightarrow \infty$. Hence, for every proposition P (the remaining clauses), the conjunction “ $d_n = O(n^{-1/2})$ and P ” is false.

Proof. If $d_n = O(n^{-1/2})$, then since $n^{-1/2} \rightarrow 0$ we get $d_n \rightarrow 0$, so $d_n < \delta$ for some $n \geq 1$, contradicting the previous theorem. $\square$

4 Lean formalization and axiom audit

File `lean/Conjecture7789/Basic.lean` (265 lines, only `import Mathlib`, Lean v4.33.1, Mathlib v4.33.1). Main statements, in namespace `Conjecture7789`:

```
theorem delta_le_worstDist (n : Nat) (hn : 1 <= n) : delta <= worstDist n
theorem worstDist_not_isBig0 :
  Not (worstDist =0[atTop] fun n : Nat => (n : Real) ^ (-(1 / 2 : Real)))
theorem conjecture_false (P : Prop) :
  Not ((worstDist =0[atTop] fun n : Nat => (n : Real) ^ (-(1 / 2 : Real))) /\ P)
```

(The Lean source writes `Nat`, `Real`, `Not`, `/\`, `<=` with their Unicode notations.) Here `delta := 1 - cdf (gaussianReal 0 1) (Real.sqrt 3)` and `delta_pos : 0 < delta`. `lake build` succeeds, and `#print axioms` reports, for each of the three theorems, `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom.